SYSTEM DESIGN

5.1 Overview of System Design

The system design of LevelFit AI is centered around creating a **highly responsive, real-time, and user-centric fitness experience**. The design ensures that every user action is processed instantly and reflected across the system with minimal latency.

The architecture follows a **layered and event-driven design approach**, where each user interaction triggers a sequence of operations across multiple system components.

5.2 High-Level System Flow (Detailed Explanation)

The complete system operates through the following pipeline:

User Interaction → Frontend Processing → API Request → Backend Logic → Database Update → AI Processing → Response Rendering

This structured flow ensures:

- Efficient request handling
- Real-time updates
- Consistent data synchronization

5.3 Detailed User Journey Flow

Step 1: User Entry Point

When the user opens the application, the system checks authentication status and routes the user to either:

- Login/Signup page
- Dashboard (if already authenticated)

Step 2: Onboarding Execution

The onboarding module collects structured user data across multiple steps:

- Fitness goals

- Body metrics
- Medical conditions
- Activity level

This data is validated and stored in the database.

Step 3: Intelligent Level Assignment

The system processes onboarding data using predefined logic rules:

- Beginner → Level 1
- Intermediate → Level 6
- Advanced → Level 11

This ensures that users start at an appropriate difficulty level.

Step 4: Daily Task Lifecycle

Each day, the system generates a set of tasks:

- Workout plan
- Diet logging requirement
- Water intake target

The user interacts with these tasks throughout the day.

Step 5: Event Trigger Mechanism

Every user action triggers an event:

- Workout completed → event fired
- Meal logged → event fired
- Water goal achieved → event fired

These events are sent to the backend for processing.

Step 6: Scoring Engine Processing

The scoring engine processes incoming events using predefined rules:

- Calculates points
- Updates streak

- Checks level progress

This is done in real-time.

Step 7: Database Synchronization

All updates are stored in the database:

- Points updated
- Progress updated
- Logs stored

This ensures data persistence and consistency.

Step 8: UI Update

The frontend fetches updated data and reflects:

- New points
- Updated progress bar
- Streak increment

This creates an instant feedback loop.

Step 9: Level Completion Logic

After completing required tasks for 5 days:

- Level is marked complete
- Bonus points awarded
- Next level unlocked

Step 10: Reset Logic Execution

If user misses tasks:

- Day 1 → notification
- Day 2 → warning
- Day 3 → reset triggered

This logic ensures accountability.

5.4 System Design Principles

1. Real-Time Feedback

Users receive immediate feedback after every action.

2. Event-Driven Architecture

System reacts dynamically to user actions.

3. Modularity

Each component (frontend, backend, AI) is independent.

4. Scalability

System can handle increasing user load.

5. Reliability

Ensures consistent performance and minimal downtime.

5.5 Data Flow Architecture

The system follows a structured data flow:

- Input: User actions
- Processing: Backend logic + scoring engine
- Storage: Database
- Output: Updated UI + AI recommendations

5.6 Error Handling & Edge Cases

Possible Issues:

- Network failure
- Incomplete data submission
- Duplicate requests

Handling Strategy:

- Retry mechanisms
- Input validation
- Idempotent API design

5.7 System Optimization Techniques

- API response caching
- Lazy loading of UI components
- Efficient database queries
- Asynchronous processing